

Emad M Meri *Mechanical Engineer*

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PROFILE

Mechanical Engineer specialized in Materials and Manufacturing with a Master’s degree from KFUPM. Solid hands-on research and academic experience. Skilled in advanced materials processing and characterization techniques. Demonstrated expertise in thermal analysis, polymer treatment, and laboratory operations.

EDUCATION

2023 – 2025	Master of Science
Dhahran KSA	<i>King Fahd University of Petroleum & Minerals (KFUPM)</i> Master of Science in Materials and Manufacturing – MSc Main courses: (Manufacturing Analysis, Polymer Engineering, Deformation of material & Design for manufacturing) Thesis topic: Carbon fiber development from alternative precursors.
2017 – 2021	Bachelor
Taif KSA	<i>Taif University</i> Bachelor of Science in Mechanical Engineering

PROFESSIONAL EXPERIENCE

Research Assistant & Lab Engineer 2023 - 2025

KFUPM

Research Contributions

- Designed and constructed experimental setups for polymer sulfonation and thermal treatment.
- Conducted advanced material characterization, including TGA, DSC, SEM, FTIR, and UTM.
- Focused on the development of carbon fiber using alternative, cost-effective precursors.
- Provided technical support & conducted material characterization tests for research projects from multiple students and research groups.

Lab responsibility

- Operated instruments & performed mechanical tests (tensile, hardness) and thermal analysis.
- Managed lab operations, including equipment maintenance, procurement, and safety compliance.
- Conducted risk assessments for chemical and thermal processes.
- Managed hazardous materials and chemical waste disposal.
- Trained students in safe laboratory practices.
- Supervised safe operation of laboratory equipment.

SKILLS

Research skills

Preparing reports, presentations, Design for experiments
Data analysis, Professional writing,

Other skills

Problem solving, Teamwork, Leadership, Business management, Engineering programs, & Microsoft Office.

Technical Skills

- Production, Processing, Machining & Additive Manufacturing.
- Design for Manufacturing (DFM).
- Quality Control, Safety Procedures & Compliance.
- Materials Engineering, Selection & characterization.

ACCREDITATION

- Registered Member, Saudi Council of Engineers (SCE).
- Certified SOLIDWORKS Professional (CSWP)

AWARDS

Active Researcher Award: KFUPM Research Forum 2024 Organized by KIKX

RESEARCH AREA

Carbon Fiber Development.
Polyethylene Curing.
Thermal Analysis of Polymer.
Material Properties.
Corrosion.
Mechanical Behavior of 3D-Printed Parts.
Hybrid & Advanced Machining.
Nanomaterials in the food industry.

THESIS

Thesis Title

EFFECTIVENESS OF SULFONATION AND NANO GRAPHENE ON THERMAL STABILITY OF POLYETHYLENE FOR THE PRODUCTION OF CARBON FIBER

Developed low-cost carbon fiber from polyethylene (LLDPE & HDPE) via sulfonation. Investigated effects of temperature, time, stress, and fiber treatments (stretching, annealing, graphene reinforcement) on thermal stability using TGA, FTIR, and SEM. Identified optimal stabilization conditions and demonstrated improved thermal and structural properties via additives.

PROJECTS

Vibration and Discharge Assisted Hybrid Electrochemical Machining

(Advanced machining course project)

Applications of Nanomaterial as Antimicrobials in Food Industry

(Nanomaterial course presentation)

Fracture of Additively Manufactured Metal Components

(Deformation of engineering material course project)

Study of the Defects in the Deep-Drawing process

(Analysis of manufacturing course project)

Polyethylene Based carbon fiber

(Engineering of Polymer course project)

Design for manufacturing of carbon fiber composites

(Design for manufacturing course project)

LSD Analysis for Heat Treatment effect on Hardness of Cobalt superalloy MAR-M50

(Applied Regression and Experimental Design course project)

PUBLICATION

Influence of Nanographene on the Sulfonation of Polyethylene for the Production of Carbon Fiber

Currently being written and will be submitted soon.

LANGUAGES

- Arabic
- English